04. Abnormal Uterine Bleeding as Classification, Not Guesswork

Companion reading handout for simultaneous listening.
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Abnormal uterine bleeding is one of the most common reasons a patient arrives in a gynecology clinic, and one of the easiest places for a resident to move too fast.

The patient says, my periods are very heavy.

And the inexperienced mind immediately reaches for a treatment. A pill. A progestin. Tranexamic acid. An intrauterine system. Maybe an ultrasound.

But heavy bleeding is not a diagnosis.

It is a doorway.

Behind that same doorway may be a teenager whose hypothalamic pituitary ovarian axis is still immature. A patient with von Willebrand disease whose first major symptom is menstruation. A thirty two year old with PCOS and unopposed estrogen. A forty eight year old with hyperplasia. A patient with a submucosal fibroid. A patient whose bleeding is actually pregnancy-related. Or a patient with a focal polyp that a blind biopsy can miss.

So the first lesson is this: abnormal uterine bleeding is not managed by instinct. It is managed by sorting.

Dominant Model

Pregnancy first. Stability second. Then classify.

After that, ask whether there is endometrial risk, structural disease, hemostatic disease, ovulatory biology, medication effect, or a fibroid whose exact location changes the entire treatment plan.

That order protects the patient from the important misses.

It protects you from calling pregnancy a period.

It protects you from treating cancer risk as inconvenience.

It protects you from reassuring a patient after a benign blind biopsy when the problem is focal.

And it protects you from treating every fibroid as if the word fibroid were enough.

Define the scope

The ACOG abnormal uterine bleeding guidance is about reproductive-aged, nonpregnant patients. It does not cover pregnancy-related bleeding, and it does not cover postmenopausal bleeding. That matters because the same bleeding description moves into a different diagnostic universe when the patient is pregnant or postmenopausal.

Normal menstrual flow generally lasts about five days. Normal cycle length is usually twenty one to thirty five days.

Older language used terms like menorrhagia, metrorrhagia, polymenorrhea, and oligomenorrhea. Those terms are still useful to understand. Menorrhagia meant heavy menstrual bleeding, classically more than eighty milliliters of blood loss. Metrorrhagia meant bleeding between periods. Polymenorrhea meant bleeding more often than every twenty one days. Oligomenorrhea meant bleeding less often than every thirty five days.

But the modern clinical language is better. We say abnormal uterine bleeding, describe the bleeding pattern, and then name the cause when we can.

And for heavy menstrual bleeding, do not become hypnotized by the eighty milliliter research threshold. In real practice, the patient's experience matters. If bleeding floods clothing, causes anemia, prevents work, wakes the patient at night, or makes normal life revolve around pads and bathrooms, then it is clinically important.

The classification language is PALM-COEIN.

Say it slowly.

PALM is structural.

P is polyp. A is adenomyosis. L is leiomyoma. M is malignancy and hyperplasia.

These are things you often need to see with imaging, see directly, sample, or remove.

COEIN is nonstructural.

C is coagulopathy. O is ovulatory dysfunction. E is endometrial. I is iatrogenic. N is not yet classified.

These are often physiologic, systemic, medication-related, or endometrial-function problems.

That split is the whole point.

PALM asks for anatomy or tissue.

COEIN asks for physiology, medications, and blood.

So when a patient says, my bleeding is heavy, your mind should not say, give hormones. It should say, which compartment is this?

Let us build the first-pass clinic pathway.

First, in any reproductive-aged patient where pregnancy is biologically possible, test for pregnancy. Do not skip this because the patient is sure the bleeding is a period. Early pregnancy loss, ectopic pregnancy, and pregnancy of unknown location can all present as bleeding. Those topics get their own lectures later in the course, but the reflex begins here.

Second, assess stability.

Is she dizzy? Syncopal? Tachycardic? Hypotensive? Passing large clots? Soaking pads rapidly? Does she look pale or unwell? The stable outpatient evaluation is different from acute heavy bleeding with hemodynamic concern.

Third, take a history that is not just menstrual dates.

You want the bleeding pattern, severity, pain, clots, flooding, anemia symptoms, contraception, medication use, anticoagulants, hormonal methods, herbal supplements, family history, personal bleeding history, pregnancy possibility, infection risk, and symptoms of endocrine disease.

ACOG specifically reminds us that medications and supplements can matter. Warfarin, heparin, nonsteroidal anti inflammatory drugs, hormonal contraception, ginkgo, ginseng, and motherwort can all enter the history.

Fourth, examine the patient in a way that matches the problem.

Look for obesity and signs of PCOS such as hirsutism and acne. Look for thyroid clues. Look for insulin resistance, such as acanthosis nigricans. Look for bleeding-disorder clues, such as petechiae, ecchymoses, pallor, or swollen joints. Their absence does not rule out a bleeding disorder, but their presence should change your level of attention.

In adults, the pelvic exam matters. Speculum exam can identify cervical, vaginal, or vulvar lesions. It can reveal discharge or cervicitis. It can show that the bleeding is not uterine at all. Bimanual exam can suggest an enlarged, irregular, firm uterus, which points you toward leiomyomas or adenomyosis.

The core laboratory evaluation is practical.

Pregnancy test.

CBC.

TSH.

Cervical cancer screening when indicated.

Consider chlamydia testing, especially when infection risk is present.

Notice what is not routine. Testosterone is not a standard first-line lab for every patient with abnormal uterine bleeding. If the history and exam suggest hyperandrogenism or PCOS, then endocrine evaluation becomes targeted. But the initial AUB workup is not a random endocrine fishing expedition.

Now pause on coagulopathy, because this is one of the most important missed diagnoses.

Up to twenty percent of women who present with heavy menstrual bleeding may have an underlying bleeding disorder. In adolescents, heavy menstrual bleeding may be the first clinical presentation of von Willebrand disease.

The history screen is structured.

A positive screen includes heavy menstrual bleeding since menarche.

Or one major bleeding clue: postpartum hemorrhage, surgery-related bleeding, or bleeding with dental work.

Or two or more smaller clues: bruising once or twice per month, epistaxis once or twice per month, frequent gum bleeding.

Or a family history of bleeding symptoms.

If the screen is positive, do not stop at a CBC. Initial testing may include platelets, prothrombin time, partial thromboplastin time, and sometimes fibrinogen or thrombin time. Bleeding time is not useful. Depending on the story, von Willebrand factor antigen, von Willebrand ristocetin cofactor activity, and factor eight testing may be needed, often with hematology involvement.

Here is the memory anchor.

If heavy bleeding started with the first periods of life, ask whether the problem is not in the uterus but in hemostasis.

Move to ovulation and endometrium

Ovulatory dysfunction is the O in COEIN.

Its classic mechanism is unopposed estrogen. If ovulation is irregular, progesterone exposure is not coordinated. The endometrium proliferates under estrogen and then sheds irregularly, sometimes heavily. This is why PCOS, thyroid disease, hyperprolactinemia, weight change, adolescence, and perimenopause can produce irregular bleeding.

Endometrial bleeding is different. In AUB E, the patient may ovulate, the axis may be intact, and steroid profiles may be normal, but local endometrial control of bleeding is abnormal. ACOG describes mechanisms such as abnormal prostaglandin synthesis, receptor upregulation, increased local fibrinolysis, and increased tissue plasminogen activator activity.

That distinction is useful because it prevents a lazy explanation.

Not every nonstructural heavy period is anovulation. Sometimes the local endometrium is the problem.

The cancer-risk question

Endometrial sampling is not for every reproductive-aged patient with abnormal bleeding. But when it is indicated, it is not optional.

For patients older than forty five with AUB, endometrial sampling is first-line.

For patients younger than forty five, sample when there is unopposed estrogen exposure, such as obesity or PCOS, when medical management has failed, or when AUB is persistent.

The mechanism is not arbitrary. Unopposed estrogen means proliferation without adequate progesterone opposition. Over time, that shifts the concern toward hyperplasia and malignancy. Age adds baseline risk. Persistence after treatment tells you that your first explanation may be incomplete.

Office endometrial biopsy is usually the first-line tissue test. But you must remember what blind biopsy can and cannot do.

It performs best when the endometrial process is global and the specimen is adequate. It is much weaker when the disease is focal. A focal polyp, a submucosal fibroid, or a focal malignancy may be missed if the sampling device does not touch it.

So make this a resident reflex.

A positive biopsy rules in disease.

A benign blind biopsy does not always rule out the problem when bleeding persists.

If the patient continues to bleed, especially with structural suspicion, move to cavity evaluation: transvaginal ultrasound, saline infusion sonography, hysteroscopy, or directed biopsy depending on the scenario.

This is a mature clinical habit. Do not turn a negative test into a false endpoint when the symptom is still telling you there is a target lesion.

Imaging

Transvaginal ultrasound is the primary uterine imaging test for reproductive-aged AUB. It is especially appropriate when the bimanual exam is abnormal, the uterus is enlarged or globular, symptoms persist despite treatment, or structural disease is suspected.

But ultrasound is a screening test, not magic.

ACOG notes that transvaginal ultrasound has limited performance for intracavitary pathology, with sensitivity and specificity in one cited comparison around fifty six percent and seventy three percent. That is why saline infusion sonography and hysteroscopy matter.

Saline infusion sonography outlines the cavity. It is better for polyps and submucosal leiomyomas, and it can distinguish focal thickening from uniform thickening.

Hysteroscopy lets you see the cavity directly and biopsy or treat in a targeted way.

MRI is not routine for AUB. It becomes useful when better mapping changes management: enlarged uterus, multiple myomas, suspected adenomyosis, planning myomectomy, planning uterine artery embolization, or when transvaginal ultrasound is inconclusive.

Do not use endometrial thickness as a reassurance shortcut in low-risk reproductive-aged AUB. In a cycling patient, thickness changes across the cycle. The postmenopausal logic does not simply transfer into reproductive age.

Now we move from abnormal bleeding in general to fibroids in particular.

The word leiomyoma appears in PALM-COEIN as AUB L. But "leiomyoma" is not enough information for treatment.

For treatment, anatomy is destiny.

Novak's fibroid chapter and the FIGO leiomyoma system give the map. Learn it from the cavity outward, not as a list of numbers.

Type zero is entirely intracavitary and pedunculated. It is in the cavity on a stalk.

Type one is submucosal with less than fifty percent intramural extension. It is mostly cavity-facing.

Type two is submucosal with fifty percent or more intramural extension. It still has a submucosal relationship, but it dives deeper into the wall.

Type three contacts the endometrium but is one hundred percent intramural. It touches the cavity lining but does not protrude into the cavity.

Type four is completely intramural.

Type five is subserosal with at least fifty percent intramural extension.

Type six is subserosal with less than fifty percent intramural extension.

Type seven is pedunculated subserosal, outside the uterus on a stalk.

Type eight is other, such as cervical, parasitic, broad-ligament, or another specified site.

Hybrid fibroids use two numbers. The first number describes the relationship to the endometrium. The second describes the relationship to the serosa. A type two dash five fibroid therefore has a submucosal component and also reaches toward the serosal side.

The classic exam trap is type five.

Type five is not submucosal. It is subserosal with at least fifty percent intramural extension.

The way to avoid the trap is spatial, not verbal. Start inside the cavity at zero. Move through the wall. Arrive at the serosa. Type five has already crossed to the outer-side family.

Ask why location matters

Submucosal fibroids distort the endometrial cavity. They are most strongly connected to heavy menstrual bleeding and impaired implantation. Novak emphasizes that submucous fibroids decrease fertility and that removing them can improve fertility.

The mechanism is anatomical and endometrial. Implantation needs a receptive, properly vascularized endometrial surface. A lesion protruding into the cavity changes the shape of that surface, disrupts local vascularity, and can alter endometrial receptivity. Novak discusses mechanisms involving endometrial receptivity pathways such as HOXA ten and HOXA eleven disruption in the setting of submucous fibroids.

Subserosal fibroids are different. They grow outward. They may cause bulk symptoms, pressure, urinary frequency, or pain if large or pedunculated, but they usually do not impair fertility. Removing a purely subserosal fibroid generally does not improve fertility.

Intramural fibroids are the gray zone. They may slightly reduce fertility, especially when large or cavity-distorting, but the benefit of removing a non-cavity-distorting intramural fibroid is uncertain. That uncertainty matters. Surgery has risks: adhesions, bleeding, anesthesia, recovery, and later pregnancy considerations.

So the resident reflex is not, fibroid equals myomectomy.

The resident reflex is, does it distort the cavity, and what is the patient's goal?

Before treatment, imaging should answer the treatment question.

Pelvic exam can suggest clinically significant intramural or subserosal fibroids when the uterus is enlarged, irregular, firm, and nontender. Ultrasound is readily available and usually the first imaging tool. But when cavity relationship will determine treatment, saline infusion sonography, hysteroscopy, or MRI may be needed.

MRI can map number, size, position, and proximity to the bladder, rectum, and endometrial cavity. Before myomectomy or uterine artery embolization, that map can change the procedure.

Treatment

Do not ask, what treats fibroids?

Ask, what is the goal?

Is the goal observation? Bleeding control? Shrinkage before surgery? Fertility preservation? Uterine preservation without current fertility? Or definitive treatment?

Many fibroids are asymptomatic. Novak emphasizes that most fibroids do not need treatment. Observation is reasonable when symptoms are absent or mild, especially near menopause, when bleeding will stop and fibroids often decrease in size.

For bleeding control, medical options include hormonal contraception, progestins, tranexamic acid, nonsteroidal anti inflammatory drugs in selected patients, and the fifty two milligram levonorgestrel IUD.

But for the IUD, anatomy returns. A levonorgestrel IUD can be excellent when the cavity is not significantly distorted. If the fibroid significantly distorts the cavity, placement and retention become problems, and the device may not treat the anatomic source well.

For endometrial polyps, the local Israeli position paper is consistent with the cavity-first logic. Diagnostic hysteroscopy is the diagnostic method of choice, and hysteroscopic polypectomy is the treatment of choice for symptomatic polyps. The point for this lecture is not to memorize every polyp pathway. It is to remember that focal intracavitary lesions are not solved by blind thinking. They need cavity-directed evaluation and often cavity-directed treatment.

For shrinkage and time buying, think about GnRH therapies.

GnRH antagonists with add-back therapy suppress gonadotropin signaling without the initial flare of agonists. ACOG's leiomyoma guidance frames oral GnRH antagonist therapy with add-back as an option for bleeding from leiomyomas for up to two years.

This is not a permanent anatomic cure. It is a way to control bleeding, correct anemia, improve quality of life, bridge to menopause, or prepare for a procedure.

GnRH agonists create a hypoestrogenic state. They can reduce bleeding, dysmenorrhea, fibroid size, and uterine size. Novak describes substantial reduction in uterine and fibroid volume with treatment, often with most shrinkage in the first months. But after discontinuation, menses returns and uterine size can return toward baseline.

So the correct memory is not, GnRH agonist cures fibroids.

The correct memory is, GnRH agonist can shrink fibroids as a short-term bridge.

Use is limited by hypoestrogenic adverse effects and bone concerns, commonly up to six months without add-back and up to twelve months with add-back in the course materials.

Fertility

For a patient who wants pregnancy now and has a cavity-distorting fibroid, myomectomy is the fertility-preserving anatomic treatment.

Hysteroscopic myomectomy is generally preferred for appropriate small submucosal fibroids. Laparoscopic, robotic, abdominal, or vaginal approaches depend on size, number, location, surgeon skill, and other pathology.

If the patient has non-cavity-distorting subserosal fibroids and infertility, myomectomy is much less likely to help fertility. If the patient has intramural fibroids, be honest about uncertainty and individualize by size, cavity relationship, symptoms, prior fertility history, and surgical risk.

Uterine artery embolization can improve bleeding and bulk symptoms in selected patients. But it is not fertility-neutral. This distinction is important.

Uterus-preserving is not the same as fertility-preserving.

UAE preserves the uterus anatomically, but Novak cautions that women who wish to conceive should not be treated with UAE because of possible ovarian damage and uncertain pregnancy effects. Some comparative data show fewer pregnancies and more miscarriages after UAE compared with myomectomy, though the evidence quality is limited. So for current fertility desire, myomectomy is usually the more conceptually appropriate anatomic intervention.

For patients who do not desire future childbearing and do not wish to retain the uterus, hysterectomy is definitive. That statement should be delivered with counseling, not reflex. The patient may value uterine preservation, recovery time, fertility, bleeding control, avoidance of repeated procedures, or certainty differently than you expect.

Radiofrequency ablation and focused ultrasound are uterus-preserving options for selected patients, but reproductive outcome data are limited, so they require careful counseling when future pregnancy is part of the plan.

Surgical Safety

Fibroids are common and sarcoma is rare. Rapid fibroid growth alone is a poor predictor of sarcoma in premenopausal women. Novak and the local morcellation paper both emphasize that rapid growth does not reliably identify sarcoma.

But the opposite error is also dangerous. We do not have a reliable way to completely exclude leiomyosarcoma before surgery.

That is why morcellation counseling matters.

Power morcellation or uncontained tissue fragmentation can disseminate an unsuspected malignancy and worsen prognosis. The local position paper stresses that preoperative assessment should include risk-factor history, physical examination, and imaging as clinically indicated, but histology after surgery may be the only definitive diagnosis.

Red flags that should change counseling and surgical planning include uterine growth after menopause, postmenopausal bleeding, intermenstrual bleeding, suspicious ultrasound or MRI features such as necrosis or rich blood flow, and elevated LDH when used in the local evaluation framework.

The teaching point is balanced.

Do not diagnose sarcoma from rapid growth alone.

Do not promise that sarcoma has been ruled out before morcellation.

Let us walk through a few cases

Case one.

A seventeen year old reports heavy bleeding since menarche. She soaks through pads, has frequent nosebleeds, and her mother says several relatives have bleeding problems.

Do not label this as immature-axis bleeding and move on. Adolescents often have anovulatory bleeding, yes. But heavy bleeding since menarche plus epistaxis and family history is a hemostasis signal. Pregnancy testing still matters if relevant. CBC matters. And the bleeding-disorder pathway matters.

Case two.

A thirty two year old has irregular heavy bleeding, obesity, acne, and hirsutism.

Now ovulatory dysfunction is likely. The mechanism is unopposed estrogen. But the same mechanism that explains the bleeding also creates endometrial risk. If bleeding is persistent, if medical management fails, or if risk is substantial, endometrial sampling becomes part of safe care even though she is younger than forty five.

Case three.

A forty eight year old has new persistent abnormal bleeding.

Do not reassure with a normal-looking ultrasound and a plan to try pills first. In AUB older than forty five, endometrial sampling is first-line. Age has moved the risk boundary.

Case four.

A forty two year old with obesity has persistent bleeding. A blind office biopsy shows benign proliferative endometrium, but bleeding continues.

The workup is not automatically over. A benign blind biopsy is strongest when the process is global. Persistent bleeding may still be a focal polyp, submucosal fibroid, or focal malignancy. Think transvaginal ultrasound, saline infusion sonography, hysteroscopy, or directed sampling.

Case five.

Ultrasound shows a type zero fibroid in a patient with heavy bleeding and anemia.

Type zero means intracavitary, usually pedunculated. The anatomy explains the bleeding. Medical therapy may stabilize or bridge, but the anatomic treatment is often hysteroscopic myomectomy when appropriate.

Case six.

A report says FIGO type five fibroid.

Do not say submucosal. Type five is subserosal with at least fifty percent intramural extension. Start at the cavity, move outward, and you will not miss it.

Case seven.

A patient wants pregnancy now and has a cavity-distorting fibroid.

An IUD does not serve her fertility goal. Combined pills may reduce bleeding but prevent conception and do not correct the cavity. UAE is uterus-preserving but not fertility-neutral. Hysterectomy is definitive but eliminates fertility. Myomectomy is the fertility-preserving anatomic treatment.

Case eight.

A patient is considering laparoscopic myomectomy and the specimen may require morcellation.

The discussion must include the benefit of minimally invasive surgery and the rare but serious risk of disseminating an occult malignancy. You cannot say that preoperative evaluation makes the risk zero. You also should not frighten the patient by implying that ordinary rapid growth equals sarcoma. The correct counseling lives between those errors.

Compress the whole lecture

Abnormal uterine bleeding is a sorting problem.

Pregnancy. Second, stability. Third, PALM-COEIN

PALM needs anatomy or tissue. COEIN needs physiology, medications, and blood.

Sample the endometrium when risk requires it: older than forty five, or younger with unopposed estrogen, failed treatment, or persistent bleeding.

Do not use a benign blind biopsy as the endpoint when bleeding persists.

Use ultrasound as the first structural screen, but use saline infusion sonography or hysteroscopy when the cavity is the question. Use MRI when mapping will change treatment.

For fibroids, locate the lesion before choosing the therapy.

Submucosal disease explains bleeding and implantation problems. Subserosal disease explains bulk more than fertility. Intramural disease requires nuance.

And remember the treatment principle.

Bleeding control, shrinkage bridge, fertility-preserving surgery, uterine preservation, and definitive treatment are different goals.

Do not let one word, fibroid, choose the treatment for you.

The resident reflex is this:

Classify the bleeding. Locate the lesion. Name the risk. Match the treatment to the patient's reproductive goal.

That is how abnormal uterine bleeding stops being a vague complaint and becomes a disciplined clinical pathway.